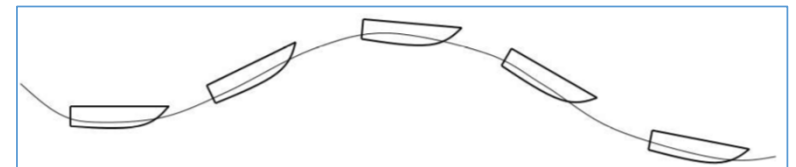


41107 Marin and ocean engineering – Lecture 10

Wave-structure interactions –
In regular waves
(Ship Motion Dynamics)

Pre-homework ... read Nielsen (2024)

- What does *coupling* between motion components mean? Why is there no coupling between heave and pitch for a ship with fore-aft symmetry? (e.g., Nielsen, 2024: “For a fore-and-aft symmetric ship at zero forward speed the cross-coupling is zero.”)* Can you explain / illustrate this.
- What does *a linear theory* refer to? E.g. (Nielsen, 2024): “All ship responses are nonlinear to some extent [with respect to waves], but in many cases when nonlinearities are small a linear theory will yield good predictions.”?
- What does it mean to consider the equations in the frequency domain? And how is this possible? What is the meaning of *encounter frequency*, e.g. Eq. (14.1) in Nielsen, 2024?
- Explain *and* draw (illustrate) why “Thus, for very long waves the heave amplitude approaches the wave amplitude and the pitch amplitude is the same as the maximum wave slope. The phasing is such that the vessel contours the waves.” (Nielsen, 2024)
- What does *added mass* (force) refer to?



*Generally, to answer this type of question, it can be helpful to ask yourself, for instance: When exerting a vertical force placed in G , will this cause a surging, a swaying, a rolling, a pitching or a yawing of the ship? (still water is assumed)

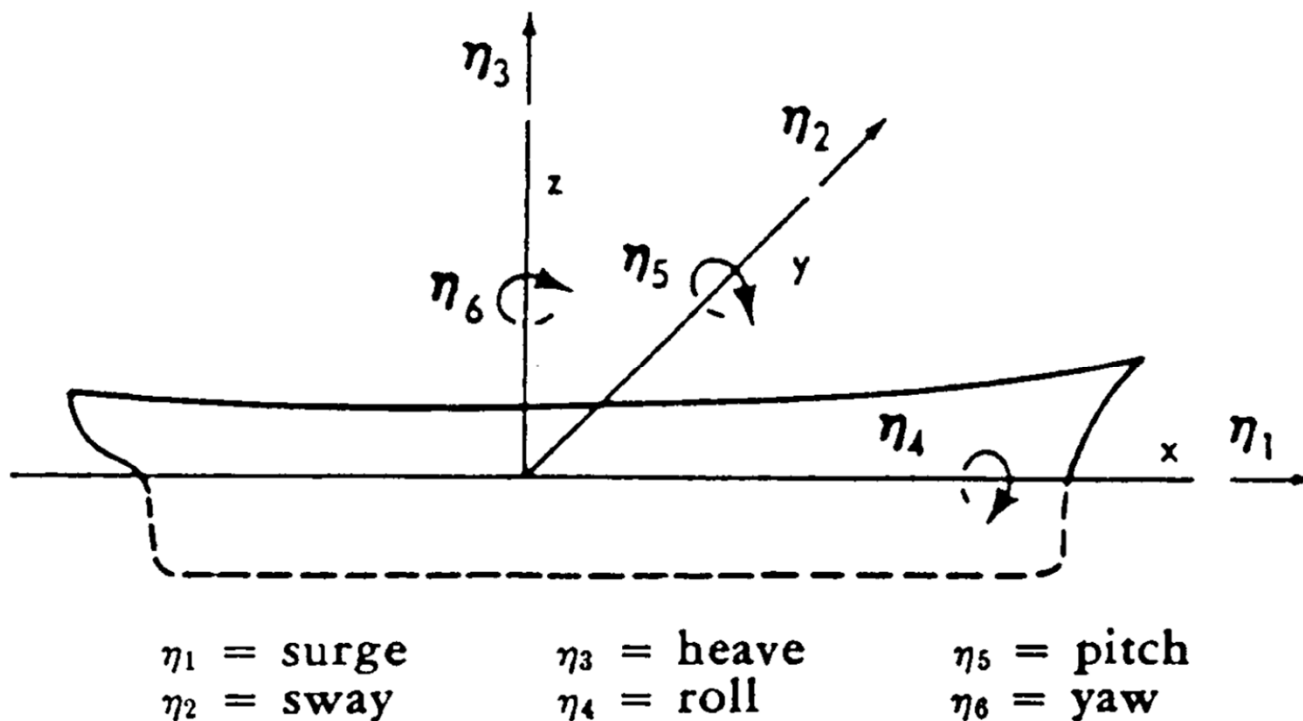
Program

- Introduction: Wave-induced ship* motions
- Forces on a floating structure in regular waves
- Transfer functions (“wave-to-motion response functions”)
- Application of the Morison formula with a semi-submersible

*Remember that *ship* is understood in the broadest sense (any floating structure)

- Reading: Chapters 14-15 in Nielsen (2024).

Ship motions – A 6 DOF problem



- Any **motion** of the ship can be made up of three translational components and three rotational components
- Knowing the resulting force on the ship, the ship's motion is governed by **Newton's 2nd law**
- In the general case, the problem at hand results in a **non-linear and coupled equation system** from which the six unknown motion components must be found...

... altogether, a highly complex problem!

Governing physics

- Newton's 2nd law for the fully coupled set of motions η :

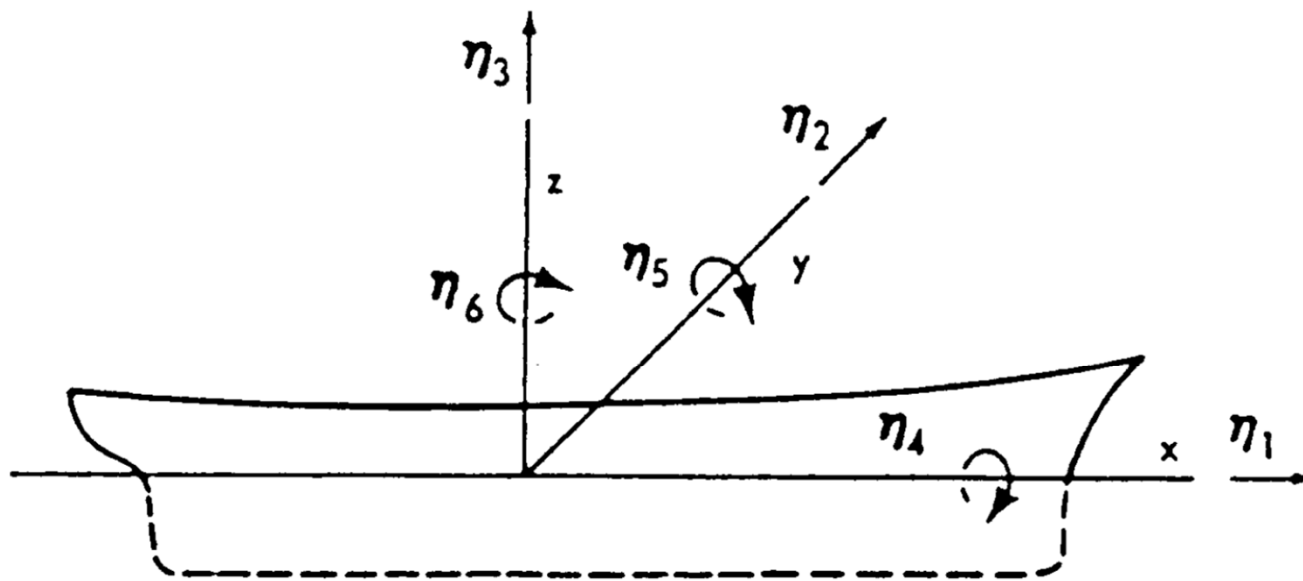
$$\sum_{k=1}^6 \Delta_{jk} \ddot{\eta}_k(t) = \mathbf{F}_j(t) \quad j = 1, 2, \dots, 6$$

- Suffice it here to note that $\mathbf{F}_j(t)$ is the *resulting* total force; emphasizing that its components are fluid forces ($j = 1, 2, 3$) *and* moments ($j = 4, 5, 6$)
- Establishing / determining $\mathbf{F}_j(t)$ is the elementary problem we are (initially) concerned about

Ship motions – Solving the problem

- Based on appropriate assumptions, **solutions** can be formulated with varying accuracy; from simple closed-form expressions to complex numerical implementations, depending on the range of assumptions
- We will *not* study the mathematical details associated with such solutions; but we shall take a **brief look at the governing physics**
- That is, we will **look at the forces** acting on a ship in waves and at the most fundamental assumptions
- Today (L10), we will limit ourselves to study a ship in *regular waves* (cf. L9); while next time (L11) we shall see the extension to irregular (ocean) waves

Ship motions – (Un)coupling



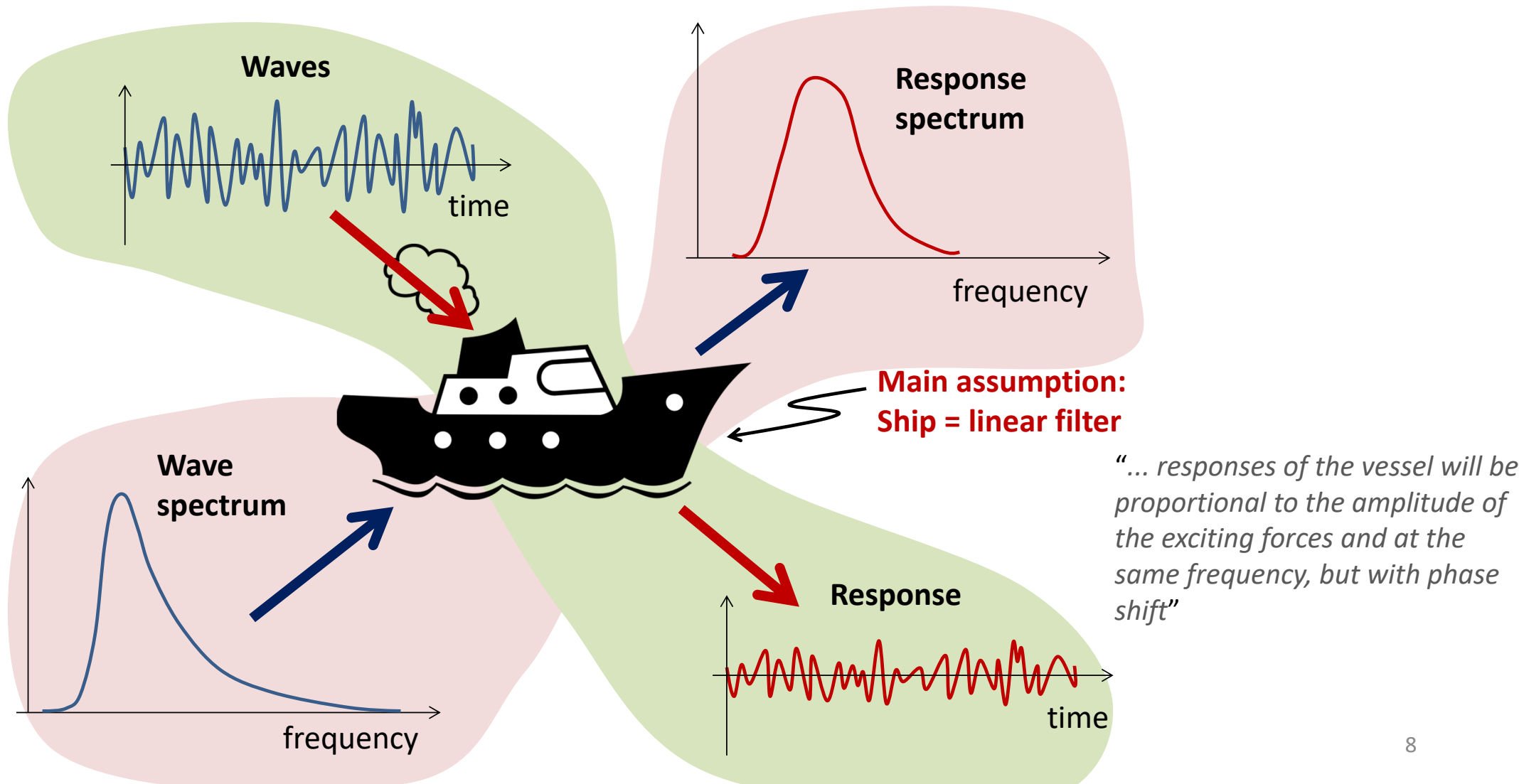
η_1 = surge
 η_2 = sway

η_3 = heave
 η_4 = roll

η_5 = pitch
 η_6 = yaw

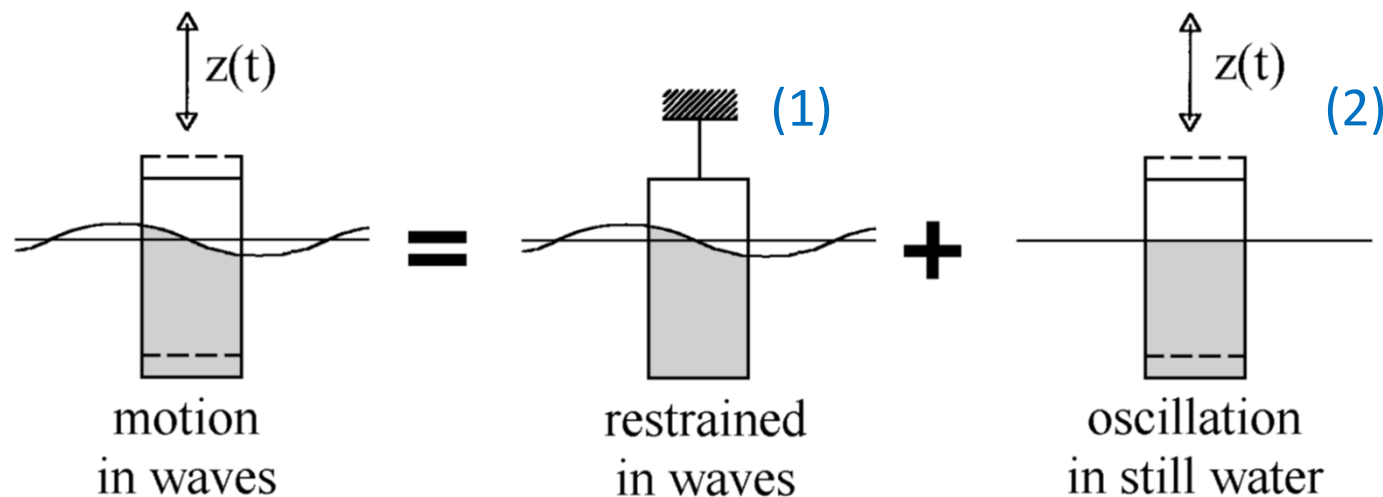
- Normally, ships have port / starboard symmetry, which means that the longitudinal-vertical plane motions (surge, heave, pitch) are *uncoupled* from the transverse-vertical plane motions (sway, roll, yaw)
- What does this actually mean?

Ship motions – Assuming linear theory

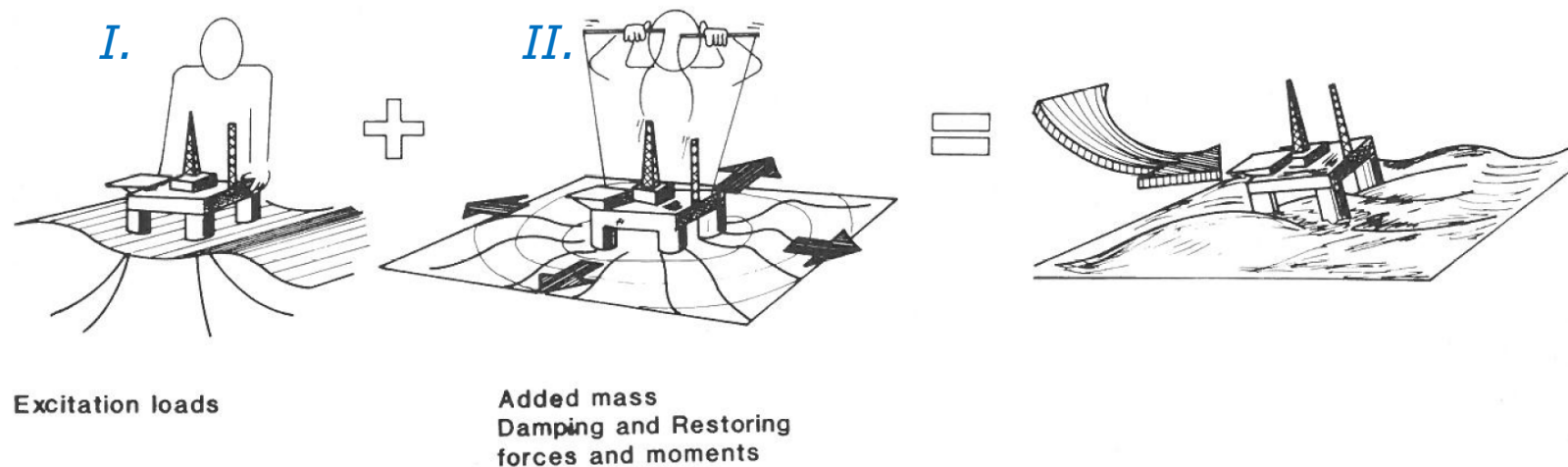


Governing physics – Excitation and radiation

- Based on the main assumption of linearity, it is possible to consider the fluid forces as the combined (= summed) result of:
 - (1) Waves acting on the restrained ship --> *Excitation*
 - (2) Movement (forced motions) of the ship in an assumed calm sea --> *Radiation*



... another way of illustrating the problem



Faltinsen, O. *Sea loads on ships and offshore structures*. Cambridge university press; 1993:

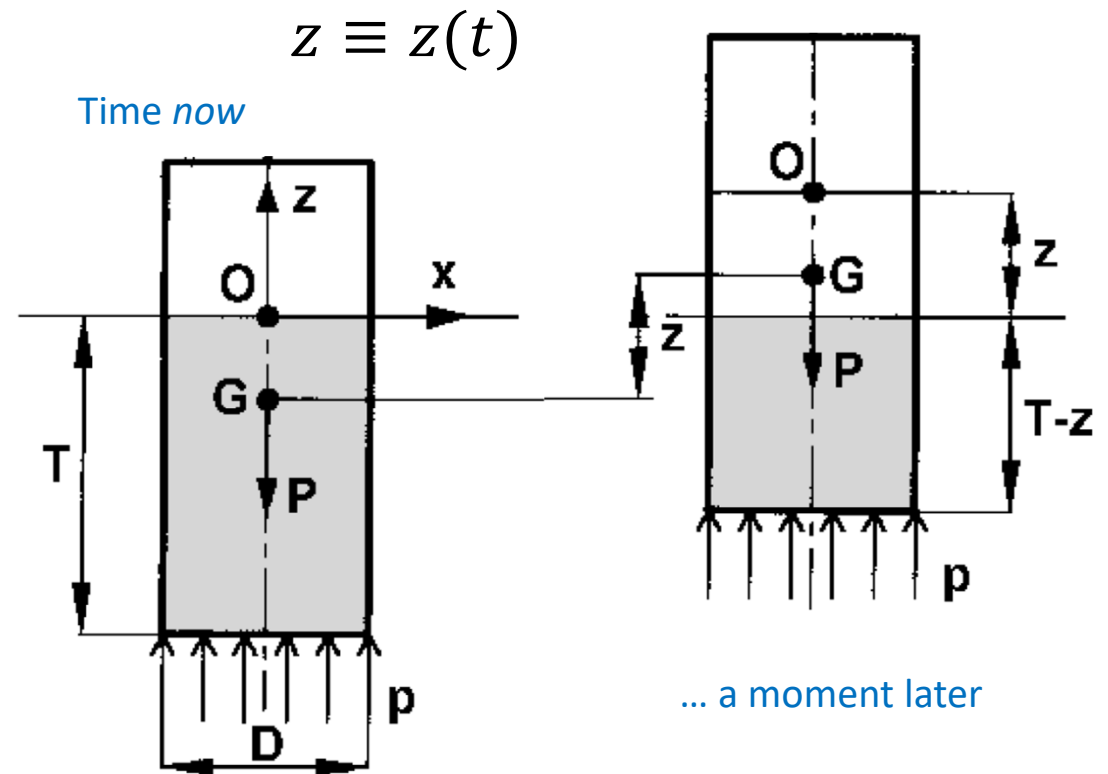
The hydrodynamic problem in regular waves is normally dealt with as two sub-problem:

I: *The forces and moments on the body when the structure is restrained from oscillating and there are incident regular waves. The hydrodynamic loads are called wave excitation loads and composed of so-called Froude-Krylov and diffraction forces and moments.*

II: *The forces and moments on the body when the structure is forced to oscillate with the wave excitation frequency in any rigid-body motion mode. There are no incident waves. The hydrodynamic loads are identified as added mass, damping and restoring terms.*

An oscillating cylinder

- Pure vertical oscillating motion around the equilibrium position $z = 0$
- Time dependency: $z \equiv z(t)$
- Motion results because of regular waves



Equation of motion

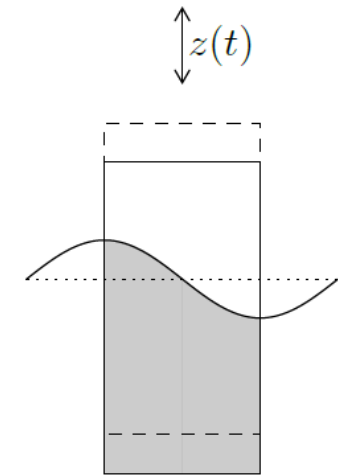
- Pure vertical motion, with $F_3(t) = F_W(t) + F_H(t)$, and $m = \rho \nabla$, the equation of motion reads

$$m \ddot{z} = F_W(t) + F_H(t)$$

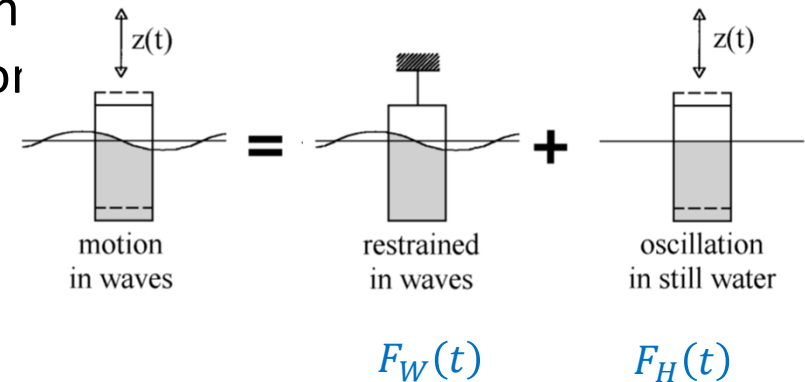
Excitation force

Hydrodynamic radiation force (for vertical motion)

- Note that the (heave) gravitational force is balanced by the buoyancy.



Motion in waves



Equation of motion

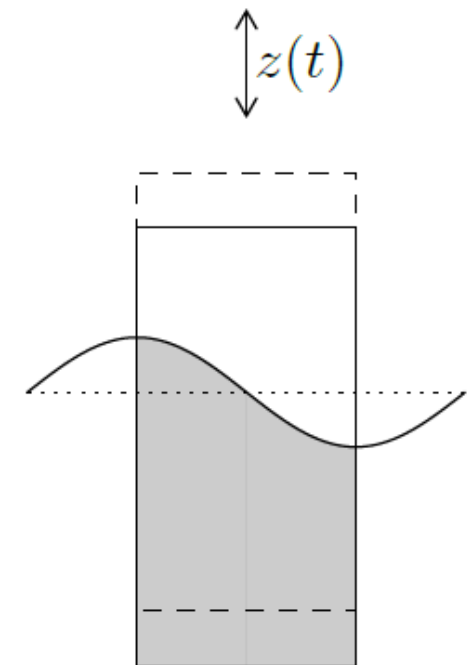
- Pure vertical motion, with $F_3(t) = F_W(t) + F_H(t)$,

$$m\ddot{z} = F_W(t) + F_H(t)$$

- In **regular / sinusoidal waves**, the exciting (heave) force is

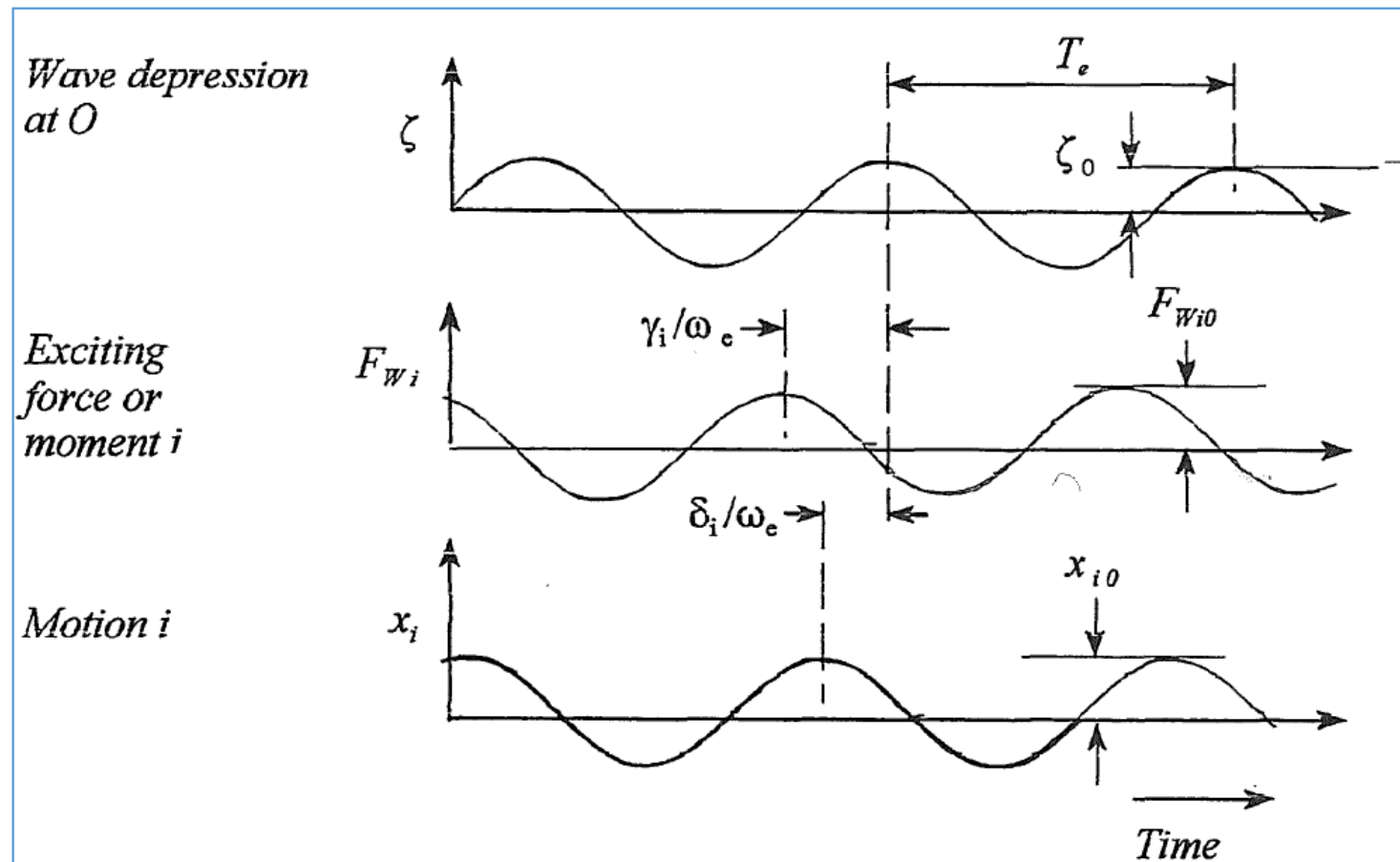
$$F_W(t) = F_a \cos(\omega t + \epsilon_{F\zeta})$$

where F_a is the amplitude of the exciting force, acting at the frequency ω , and $\epsilon_{F\zeta}$ is the phase relative to the wave.



Motion in
waves

The BIG picture – Conceptual illustration



NOTE: Different symbols!

Radiation force

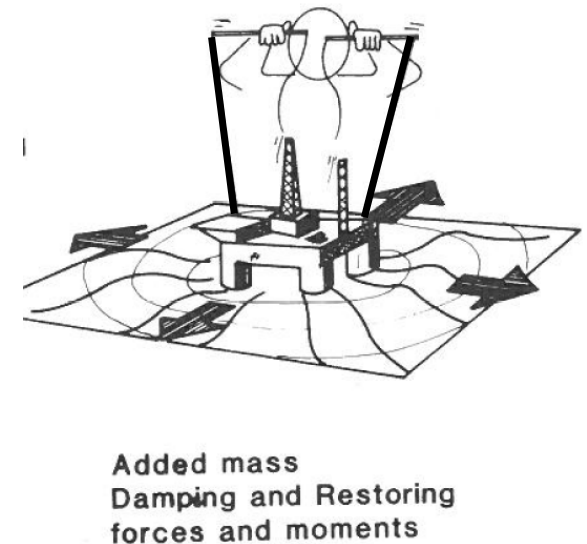
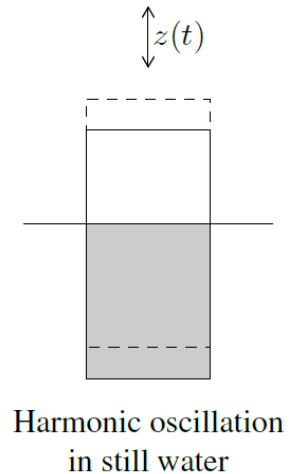
- The *hydrodynamic radiation force* in otherwise calm water is expressed in terms assumed to be proportional to vertical displacement, velocity, and acceleration

$$F_H(t) = -(a\ddot{z} + b\dot{z} + cz)$$

$a \equiv a(\omega)$: Added mass coefficient

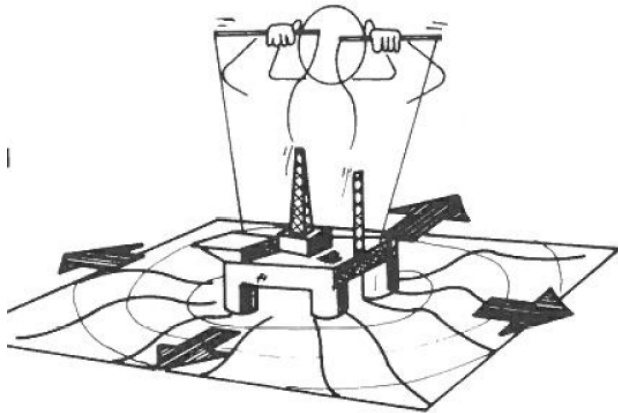
$b \equiv b(\omega)$: Hydrodynamic damping coefficient

c : Restoring (“spring”) coefficient



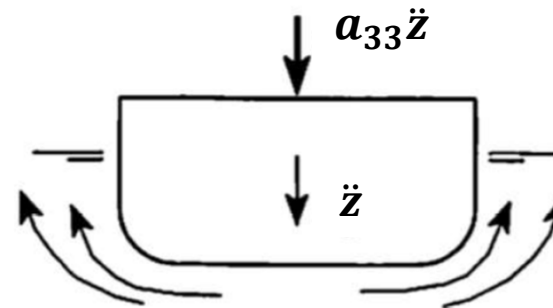
Homework: Is it reasonable to assume that added mass and damping depend on the wave (oscillating) frequency? Will they increase/decrease with frequency?

The *Radiation Force*... can itself be decomposed into three contributions



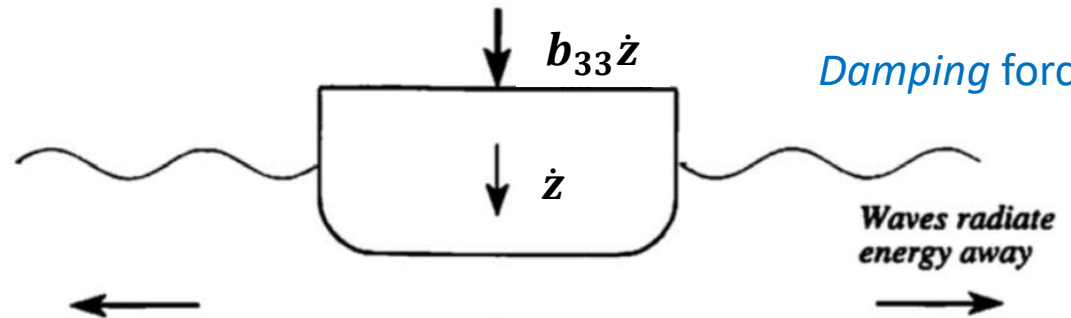
Added mass
Damping and Restoring
forces and moments

(a) Heave acceleration



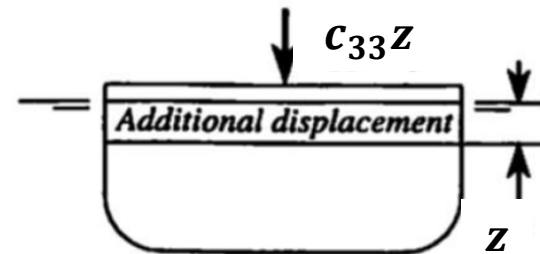
Added mass force

(b) Heave velocity



Damping force

(c) Heave displacement



Restoring force

An oscillating cylinder – Revisiting the Equation of Motion

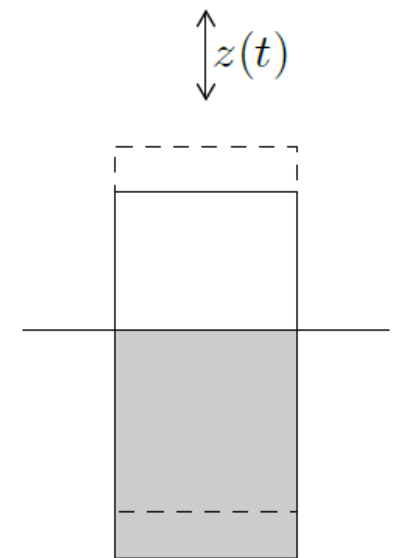
- The *hydrodynamic radiation force* in otherwise calm water is expressed in terms assumed to be proportional to vertical displacement, velocity, and acceleration

$$F_H(t) = -(a\ddot{z} + b\dot{z} + cz)$$

- Thus, **the linear equation of motion reads**

$$m\ddot{z} = F_W(t) + F_H(t) \Rightarrow$$

$$(m + a)\ddot{z} + b\dot{z} + cz = F_a \cos(\omega t + \epsilon_F \zeta)$$



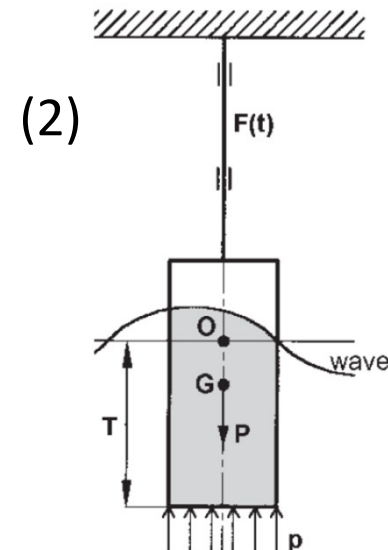
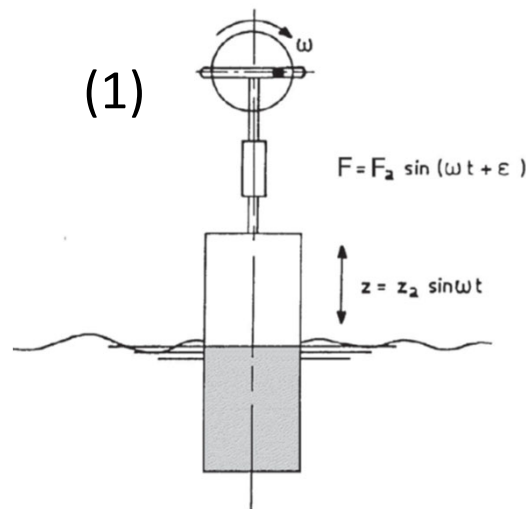
Harmonic oscillation
in still water

Example: Harmonic motion and the *natural* frequency

1. Consider the motion described by $x(t) = x_0 \sin(\omega t + \epsilon)$.
 - Determine the velocity and the acceleration; and specify the amplitudes.
2. The motion x is governed by a spring-mass-damper system with effective mass $m + a$, damping coefficient b , and stiffness coefficient c . The system forcing (function) can be described by $F(t) = F_0 \sin(\omega t)$.
 - Write the equation of motion for this system.
 - What is the natural *undamped* frequency of the system (i.e. $b = 0$)?
 - What is the difference between the natural frequency of the system and the exciting frequency (ω)?

Hydrodynamic coefficients

- It appears that the main complication in solving the hydrodynamic problem of the oscillating cylinder is **the determination of a , b , and c** ; and it is!
- It is at this point that the problem is decomposed into those of (1) a cylinder making *forced oscillations in otherwise still water* and (2) a cylinder being *restrained but excited by waves*



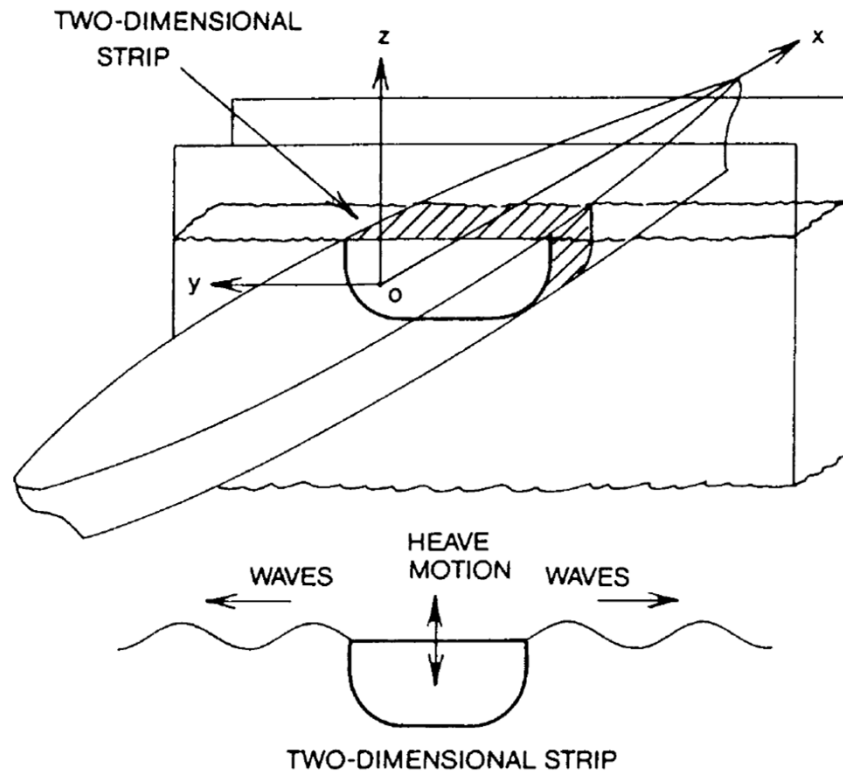
Hydrodynamic coefficients

- The details (and complexities) involved in the determination of the hydrodynamic coefficients ***a***, ***b***, ***and c*** for *arbitrary geometries* (e.g. ships) are beyond our scope.
- Generally, the same technique is applied; but closed-form* solutions cannot be derived...

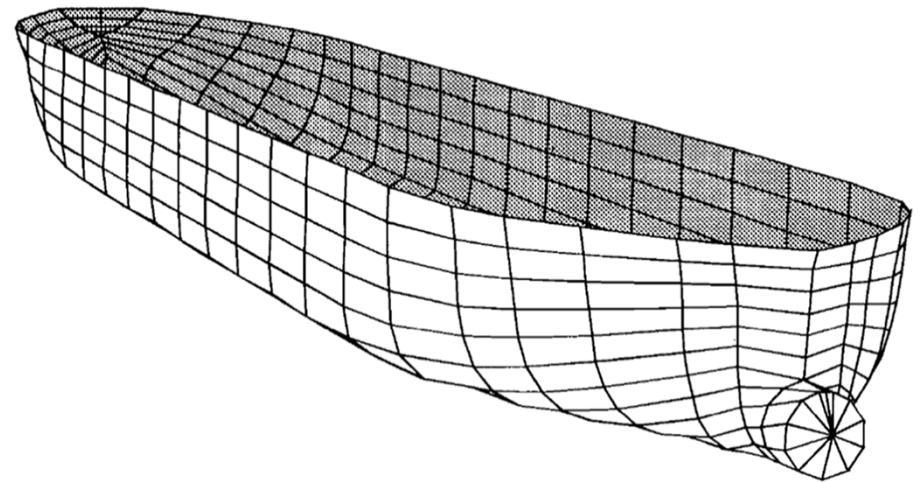
*Note, however, that for the special case of a box-shaped vessel, Jensen et al. (2004) have derived closed-form expressions for the estimation of (uncoupled) heave and pitch. Available on Learn]

Calculation of excitation and added mass using *potential theory*

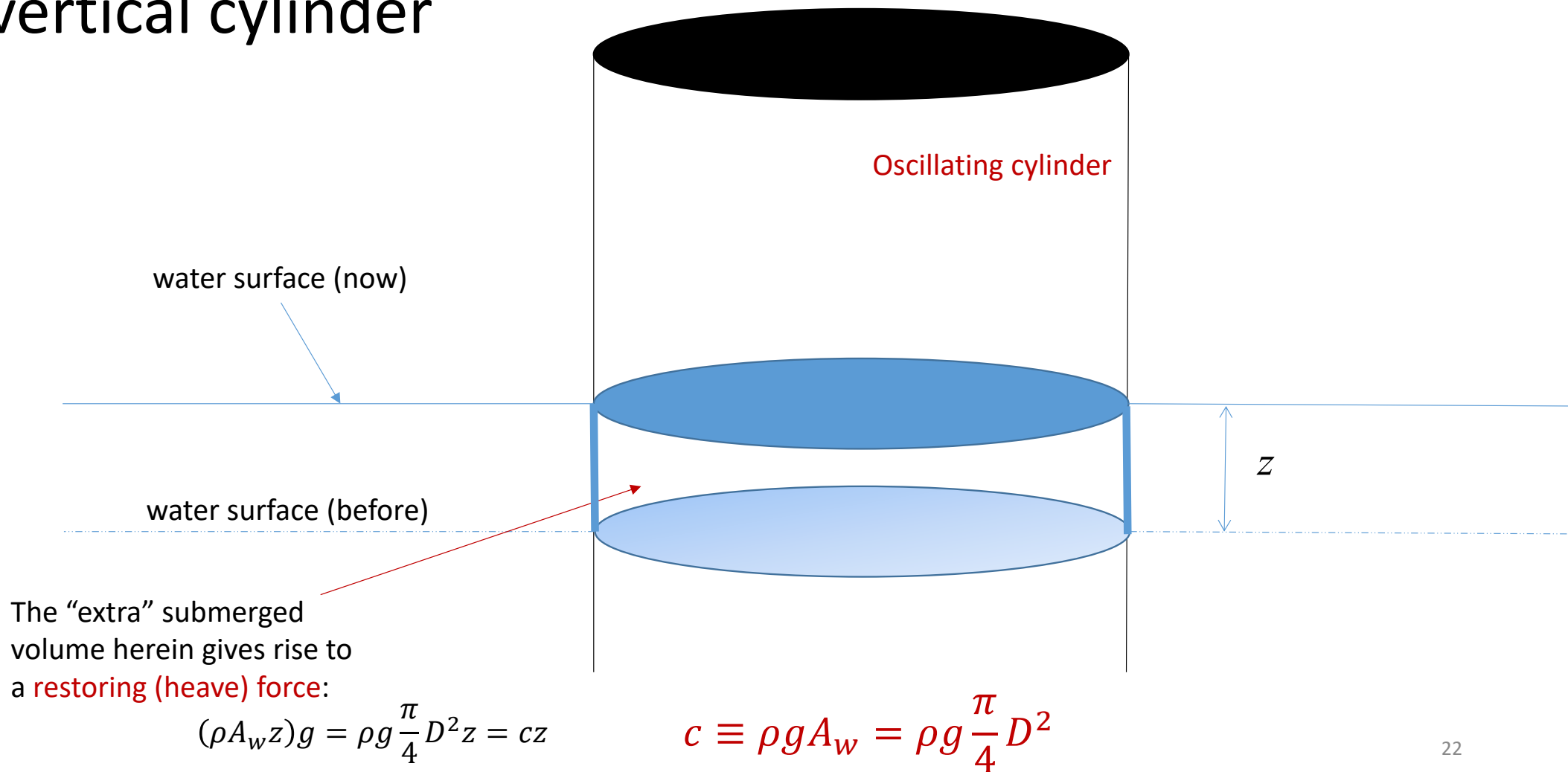
Two-dimensional strip method



Three-dimensional panel method

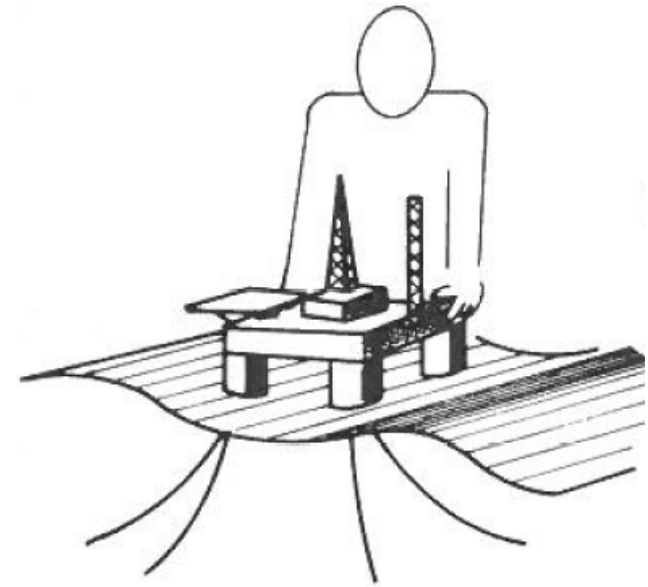


The restoring coefficient c for a vertical cylinder



Excitation force on a cylinder

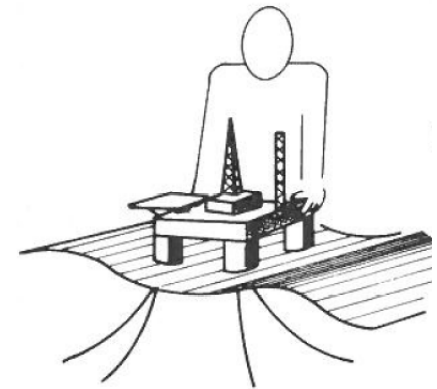
- The excitation force results because of the dynamic **pressure variation** and due to **diffraction**



Excitation loads

Excitation force on a cylinder

- The excitation force: Froude-Krylov (pressure variation) + diffraction excitations, **treated earlier in L4 and L5** (chapters 4 and 6 in Sumer and Fredsøe, 2006)
- From the lecture notes (Nielsen, 2024):
 - The **Froude-Krylov excitations** represent the integration of the pressure over the body surface that would exist in the incident wave system if the body were not present.
 - The **diffraction excitations** are caused by the diffraction or modification of the incident waves due to the presence of the vessel.



Excitation force – Restrained cylinder

The vertical cylinder is restrained and, at the same time acted upon by waves:

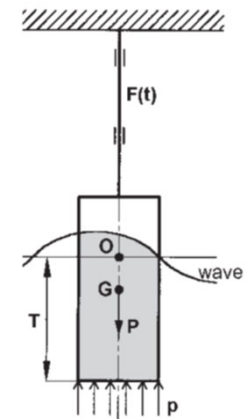
- **Froude-Krylov force:** Recall (L9) that the dynamic pressure in regular waves is

$$p = \rho g \zeta_a e^{kz} \cos(\omega t - kx)$$

- Thus, the vertical Froude-Krylov force on the bottom of a *small* cylinder (relative to wave length)

$$F_{FK} = A_s p_s = A_s \rho g \zeta_a e^{kT} \cos(\omega t) \equiv c \zeta^*,$$

where $A_s = A_w$ is the bottom area of cylinder, and T its “mean” draught.



$$\zeta^* = \zeta_a e^{-kT} \cos \omega t$$

Reduced wave elevation,
or effective wave elevation

Excitation force – Restrained cylinder

Diffraction force

- A part of the waves will be diffracted, requiring a correction to the Froude-Krylov force. The correction consists of two components:
 1. one proportional to the vertical **velocity** of the water particles and $w = \zeta_a \omega e^{kz} \sin(kx - \omega t)$
 2. one proportional to the vertical **acceleration** of the water particles $\dot{w} = -\zeta_a \omega^2 e^{kz} \cos(kx - \omega t)$
- That is, diffraction force is also expressed in terms of the reduced wave elevation ζ^* , since $w = \dot{\zeta}^*$ and $\dot{w} = \ddot{\zeta}^*$

Total force: Froude-Krylov + diffraction

$$F_W = a\ddot{\zeta}^* + b\dot{\zeta}^* + c\zeta^*$$

$$\zeta^* = \zeta_a e^{-kT} \cos \omega t$$

$$\dot{\zeta}^* = -\zeta_a e^{-kT} \omega \sin \omega t$$

$$\ddot{\zeta}^* = -\zeta_a e^{-kT} \omega^2 \cos \omega t$$

Excitation force – Restrained cylinder

Insertion... gives the *total* excitation force

$$F_W = \zeta_a e^{-kT} (c - a\omega^2) \cos \omega t - \zeta_a e^{-kT} (b\omega) \sin \omega t$$

and we also have

$$\begin{aligned} F_W &= F_a \cos(\omega t + \varepsilon_{F\zeta}) \\ &= F_a \cos \varepsilon_{F\zeta} \cos \omega t - F_a \sin \varepsilon_{F\zeta} \sin \omega t \end{aligned}$$

$$F_a \cos \varepsilon_{F\zeta} = \zeta_a e^{-kT} (c - a\omega^2)$$

$$F_a \sin \varepsilon_{F\zeta} = \zeta_a e^{-kT} (b\omega)$$

Equating cosine and sine terms yields the amplitude and phase of the excitation force:

$$\frac{F_a}{\zeta_a} = e^{-kT} \sqrt{(c - a\omega^2)^2 + (b\omega)^2}$$

$$\varepsilon_{F\zeta} = \arctan \left(\frac{b\omega}{c - a\omega^2} \right)$$

Transfer functions

- Determination of the response as function of the frequency of the waves....

Response in regular waves – Transfer function

- The central assumption in the linear theory is that the *“the harmonic responses of the vessel will be proportional to the amplitude of the exciting forces and at the same frequency, but with phase shift”*
- For the cylinder, this means

$$\begin{aligned}z &= z_a \cos(\omega t + \epsilon_{z\zeta}) \\ \dot{z} &= -z_a \omega \sin(\omega t + \epsilon_{z\zeta}) \\ \ddot{z} &= -z_a \omega^2 \cos(\omega t + \epsilon_{z\zeta})\end{aligned}$$

- **with z_a proportional to the wave amplitude**, but we expect a dependency with wave frequency. Therefore **$z_a(\omega) = |\Phi(\omega)| \zeta_a$** .

Response in regular waves – Transfer function

- Substitution of the (“unknown”) solution into the governing equation:

$$z_a \{c - \omega^2(m + a)\} \cos(\omega t + \epsilon_{z\zeta}) - z_a b \omega \sin(\omega t + \epsilon_{z\zeta}) \\ = F_a \cos(\omega t + \epsilon_{F\zeta})$$

- Inserting amplitude and phase of excitation force (slide 27), leads to:

$$\frac{z_a}{\zeta_a} = e^{-kT} \sqrt{\frac{(c - a\omega^2)^2 + (b\omega)^2}{(c - (m + a)\omega^2)^2 + (b\omega)^2}}$$
$$\epsilon_{z\zeta} = \arctan \left(\frac{-mb\omega^3}{(c - a\omega^2)(c - (m + a)\omega^2) + (b\omega)^2} \right)$$

Homework: Check the equations.

Response in regular waves – Transfer function

- In combination, the normalized response amplitude $\frac{z_a}{\zeta_a} \equiv |\Phi(\omega)|$ and phase angle $\epsilon_{z\zeta}$ constitute the **transfer function**.
- Note that, sometimes, we think of the transfer function just in terms of the magnitude $|\Phi(\omega)|$ (and as usual, the devil goes by many names... *frequency response function*, *response amplitude operator*,)

Transfer function expressed via complex notation

- Write the response in complex notation (unit amplitude waves are assumed for convenience):

$$\bar{z} = \zeta_a \bar{\Phi}(\omega) e^{i\omega t} = \zeta_a \bar{\Phi}(\omega) \{\cos(\omega t) + i \sin(\omega t)\}$$

where the complex amplitude is

$$\bar{\Phi}(\omega) = \text{Re}[\bar{\Phi}(\omega)] + i \text{Im}[\bar{\Phi}(\omega)] = \bar{\Phi}_{Re} + i \bar{\Phi}_{Im}$$

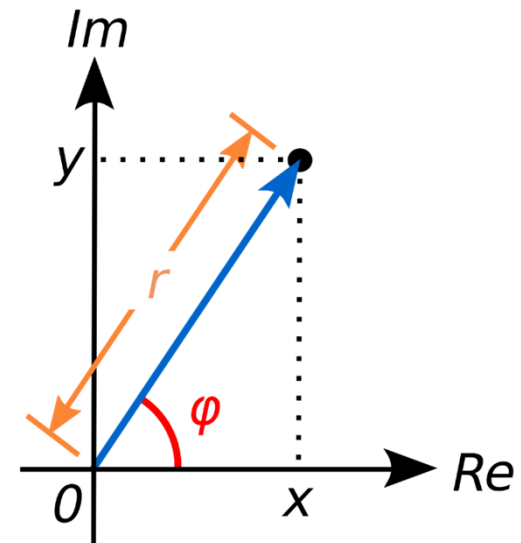
- In this case,

Response amplitude:

$$\Phi(\omega) \equiv |\bar{\Phi}(\omega)| = \sqrt{\bar{\Phi}_{Re}^2 + \bar{\Phi}_{Im}^2}$$

Phase angle (relative to wave):

$$\tan \epsilon_{z\zeta} = \frac{\bar{\Phi}_{Im}}{\bar{\Phi}_{Re}}$$



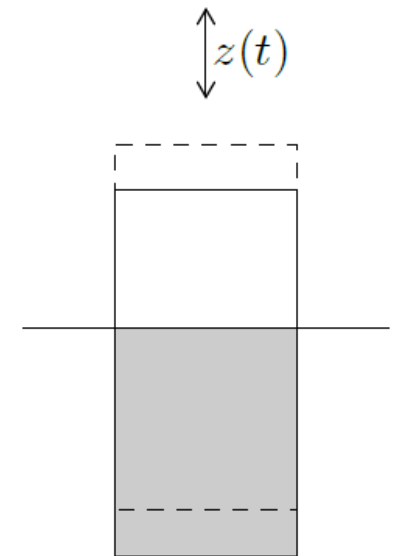
A simplified example: An oscillating cylinder... revisited

- Linear equation of motion reads

$$(m + a)\ddot{z} + b\dot{z} + cz = F_a \cos(\omega t + \epsilon_{Fz})$$

- Note that the **undamped** natural frequency is given by

$$\omega_n = \sqrt{\frac{c}{m + a}}$$



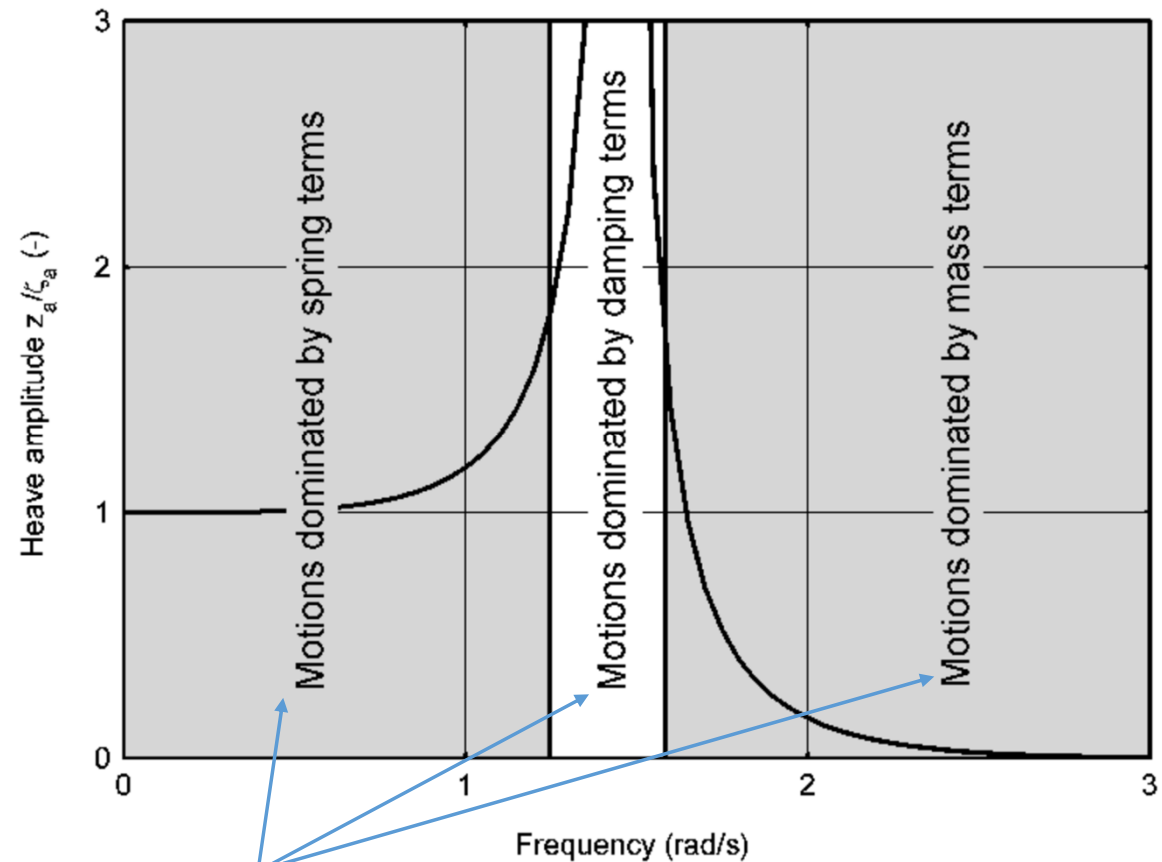
Harmonic oscillation
in still water

Question: How do we derive this result?
(see earlier example...)

Motion response of the oscillating cylinder – A numerical example

- $$|\Phi(\omega)| = \frac{F_a}{\zeta_a \sqrt{\{c - \omega^2(m+a)\}^2 + b^2 \omega^2}}$$

$$= \frac{e^{-kT} \sqrt{(c - a\omega^2)^2 + b^2 \omega^2}}{\sqrt{\{c - \omega^2(m+a)\}^2 + b^2 \omega^2}}$$
- Assume cylinder geometry; this gives a and c , and further we study for $b = 0$. Select a range of ω ...



How/why can we say this? (ans.: consider above expression)

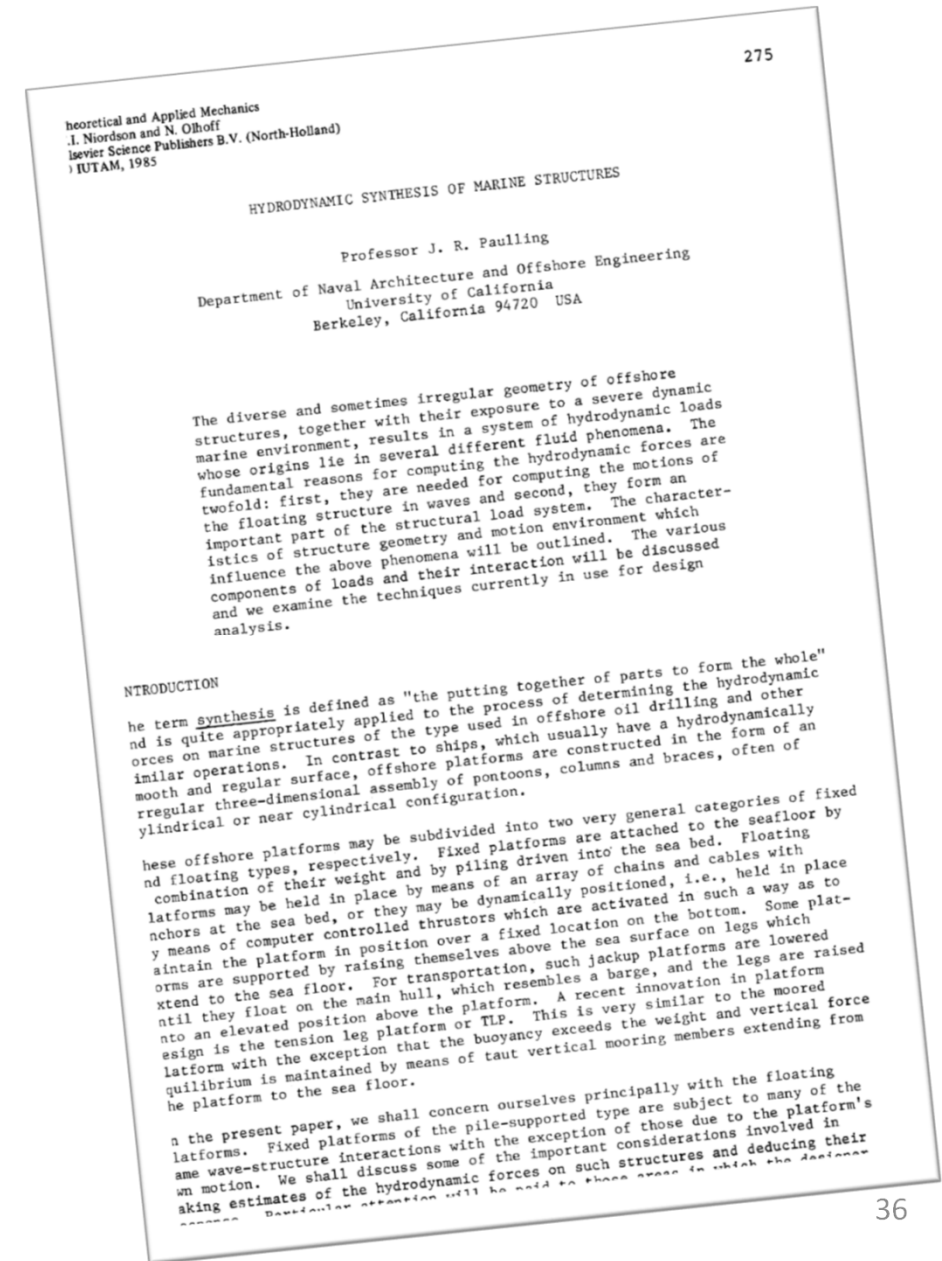
The Morison formula applied with a semi-submersible (Paulling, 1985 → Nielsen, 2024)



Not presented

Homework related to the assignment

- What is the advantage that (Paulling, 1985, p. 276): *"the immersed volume is deeply submerged and the waterplane area is relatively small in relation to the total volume."*
- Discuss also the other two points; what are their consequence from a hydrodynamical perspective?:
 1. *"Most of the tubular members are relatively small in cross section compared to their own length and to the length of important wave components."*
 2. *The spacing between members is large compared to the member cross section."*
- How has Eq. (10) in Paulling (1985, p. 281) been derived? Try to do this.



Morison formula and cylindrical members

- Recall the Morison formula; e.g., Eq. (4.26) in Sumer and Fredsøe (2006), see L4:

$$q(t) = \frac{1}{2}\rho C_D U|U| + \rho A \dot{U} + \rho C_m A_R \dot{U}$$

Drag force
Froude-Krylov force
Hydrodynamic (“added mass”) force

q : In-line force per unit length of cylinder

t : Time

U : Particle velocity (perpendicular to cylinder axis)

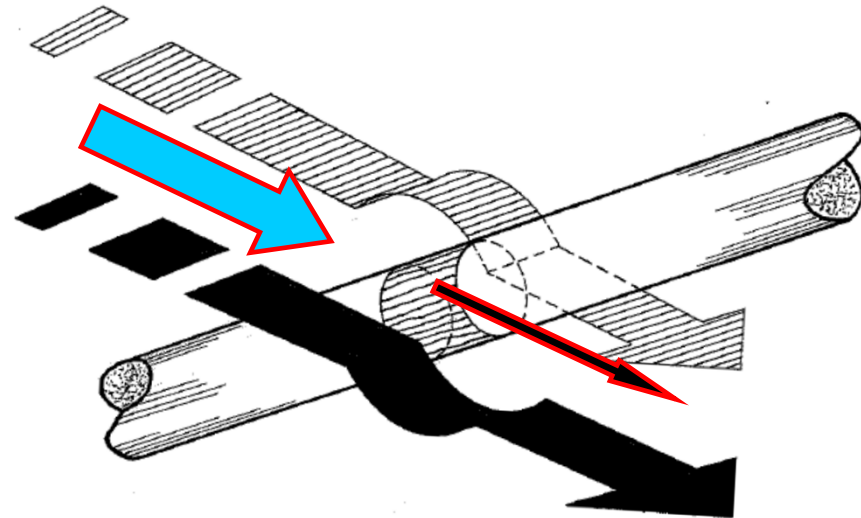
$\dot{U}$: Particle acceleration

C_D : Drag coefficient

C_m : Hydrodynamic mass coefficient (added water)

A : Cross section area (cylinder)

A_R : Reference area (added water)

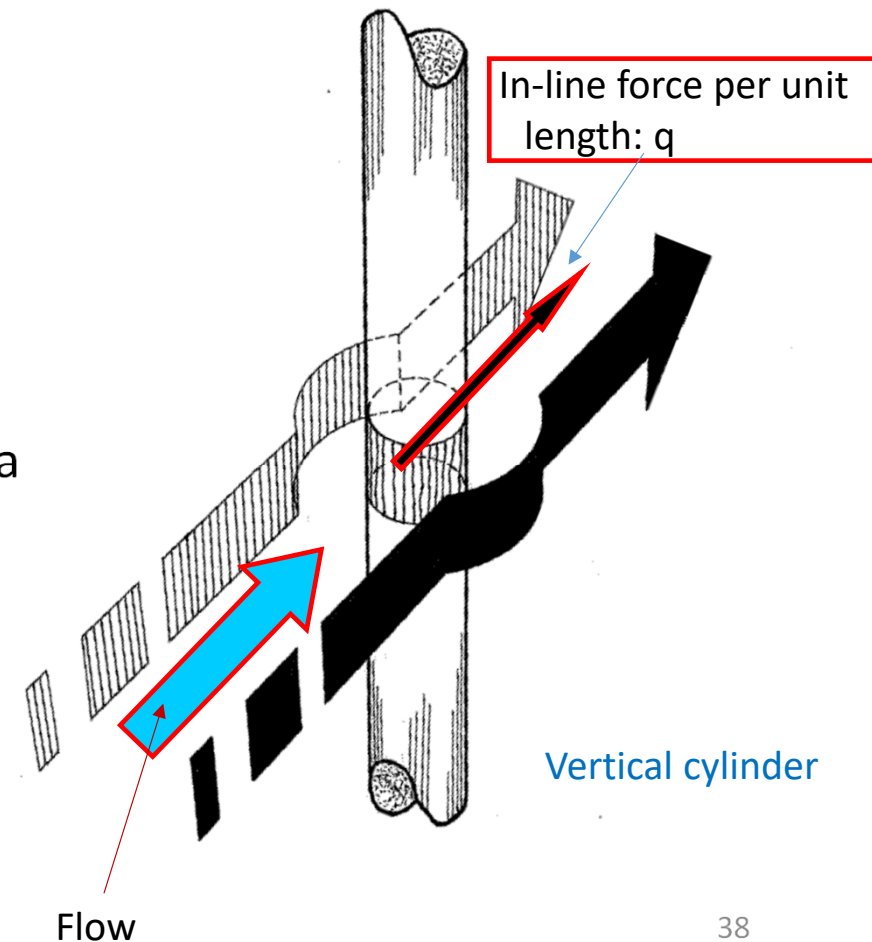


Morison formula and cylindrical members

- Morison formula:

$$q(z, t) = \frac{1}{2} \rho C_D U |U| + \rho A \dot{U} + \rho C_m A_R \dot{U}$$

- In the following, we will apply the Morison formula to slender cylindrical members aligned both vertically and horizontally
- The flow (water particles) is a result of regular waves; there will be a **horizontal component** dependent on depth z...



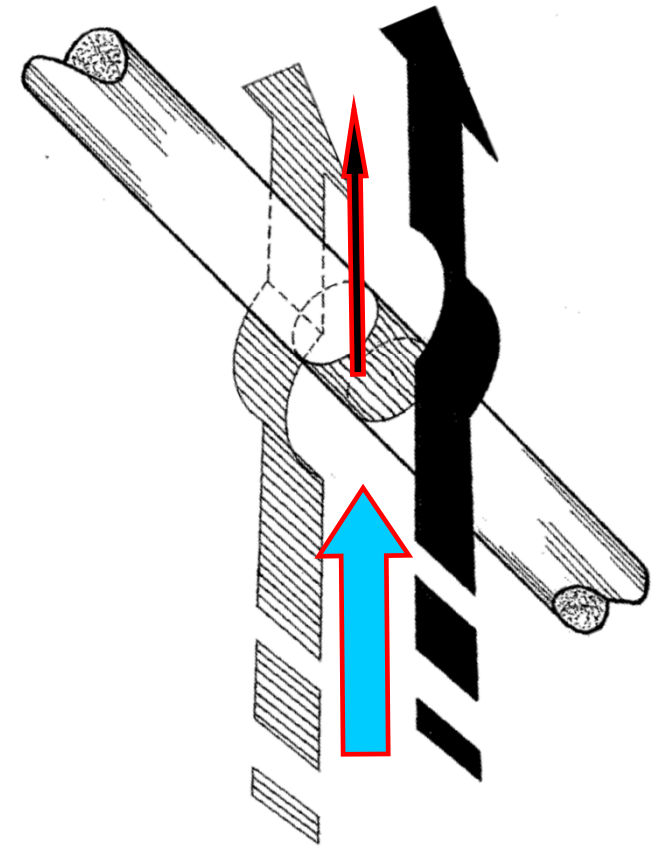
Morison formula and cylindrical members

Horizontal cylinder

- Recall the Morison formula; e.g., Eq. (4.26) in Sumer and Fredsøe (2006), see L4:

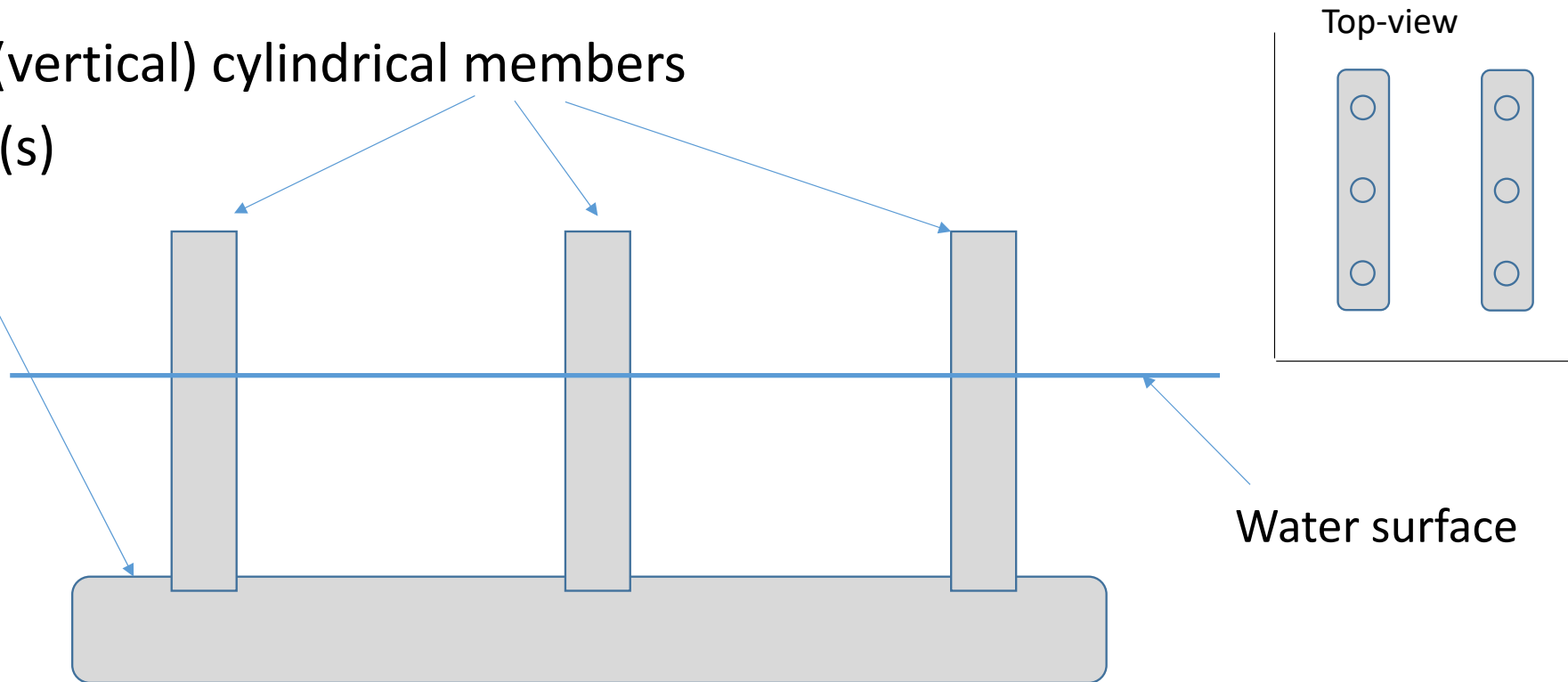
$$q(z, t) = \frac{1}{2} \rho C_D U |U| + \rho A \dot{U} + \rho C_m A_R \dot{U}$$

- ... but there can also be a **vertical component**; in this case shown together with a horizontally aligned cylinder



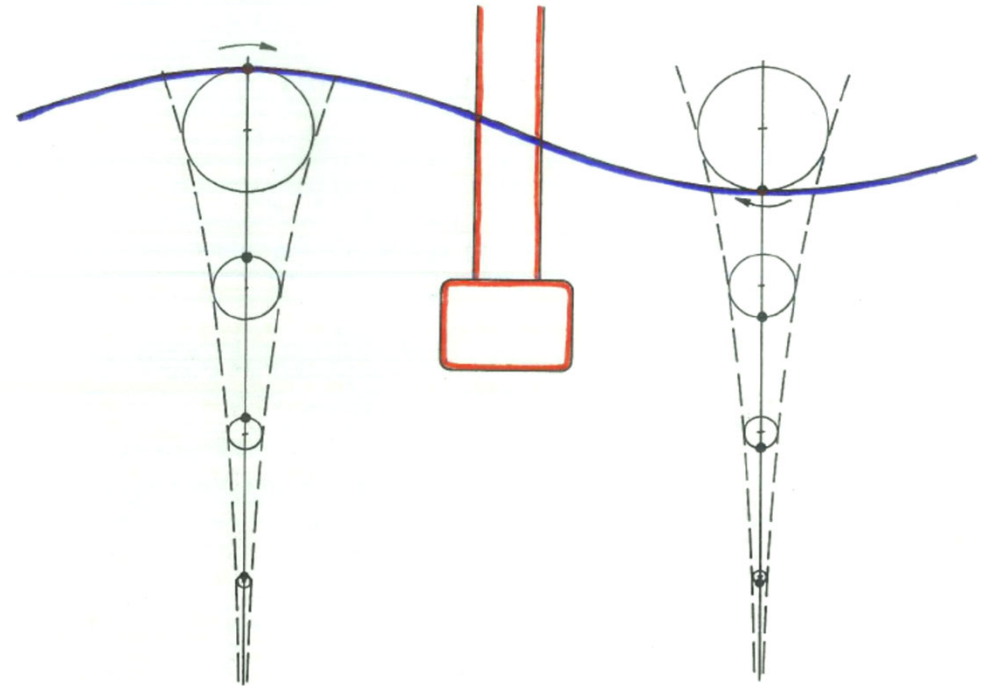
Semi-submersible – Conceptual illustration

- Slender (vertical) cylindrical members
- Pontoon(s)



Semi-submersible – The advantage

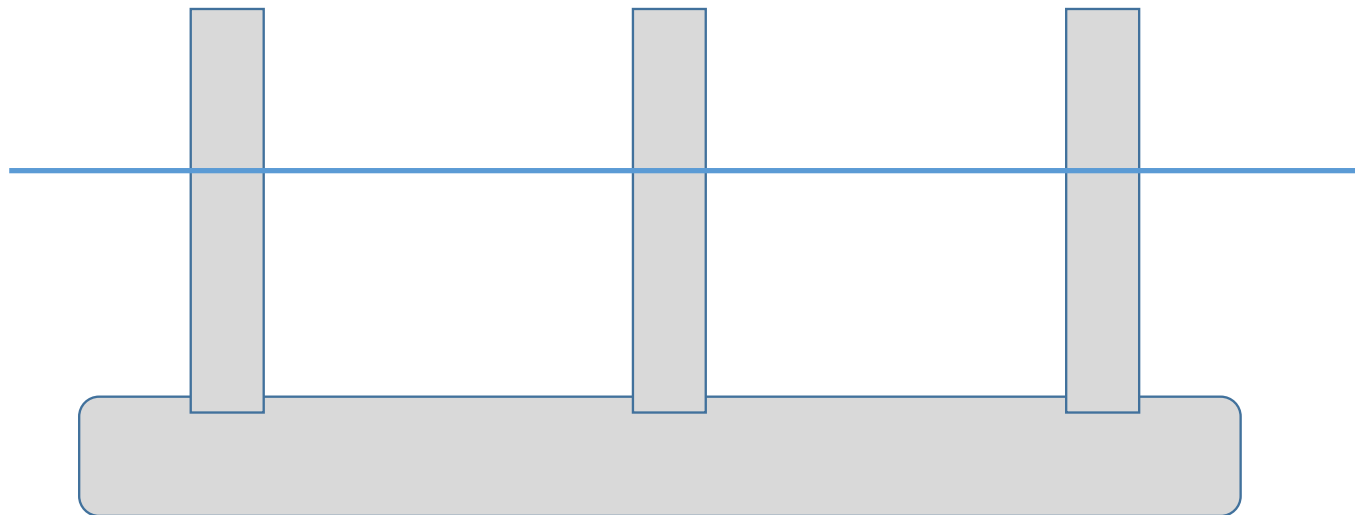
- Small water surface area
- Displacement (buoyancy) below the sea surface



A simplified analysis of a semi-submersible

- In the *hydrodynamic synthesis* (Paulling, 1985) we assume:
 - slender (relative to wavelength) cylindrical members of surface piercing elements; i.e. wave diffraction can be neglected
 - no hydrodynamic interaction between the single structural elements comprising the “exploded” platform...

Semi-submersible... “exploded”



A simplified analysis of a semi-submersible

- In the *hydrodynamic synthesis* (Paulling, 1985) we assume:
 - slender (relative to wave length) cylindrical members of surface piercing elements; i.e. wave diffraction can be neglected
 - no hydrodynamic interaction between the single structural elements comprising the “exploded” platform
 - small amplitude waves; i.e. linear theory applies
 - Morison formula holds for the calculation of the wave loads (= excitation force); i.e. of the “in-line” force on fully submerged member
- NB: The wave loads (using Morison formula) are calculated with *no* account to the absolute motion of the platform

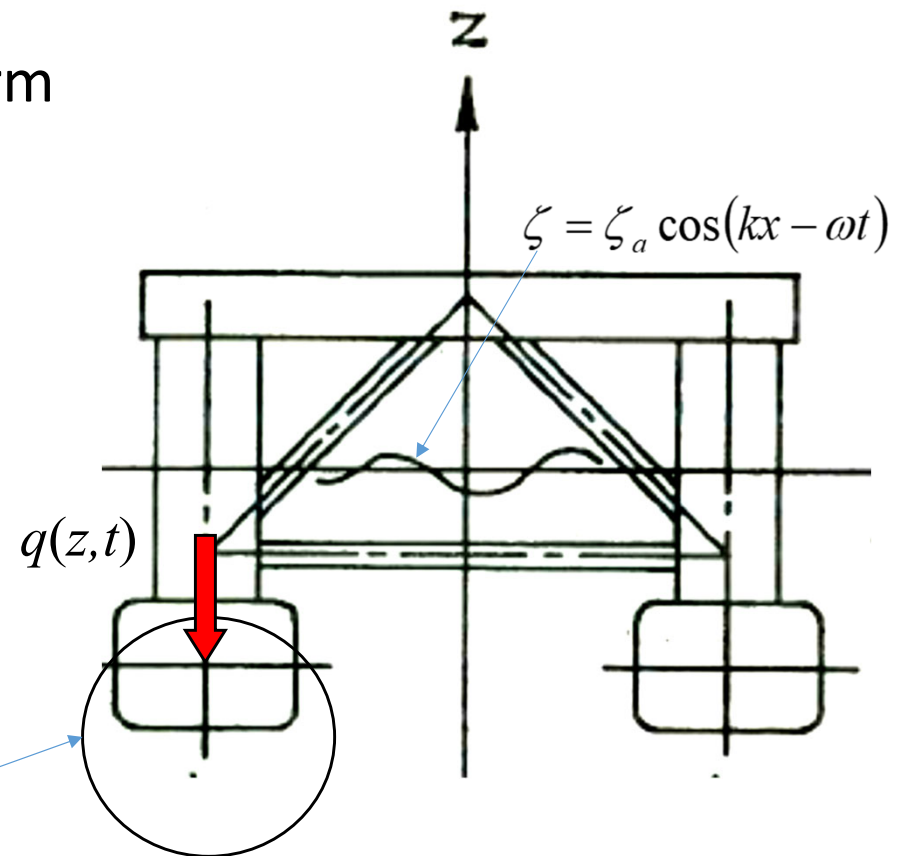
A simplified analysis of a semi-submersible, cont'd

- Morison formula without damping term

$$q(z, t) = \rho A \dot{U} + \rho C_m A_R \dot{U}$$

$$u = \zeta_a \omega e^{kz} \cos(kx - \omega t)$$

$$w = \zeta_a \omega e^{kz} \sin(kx - \omega t)$$

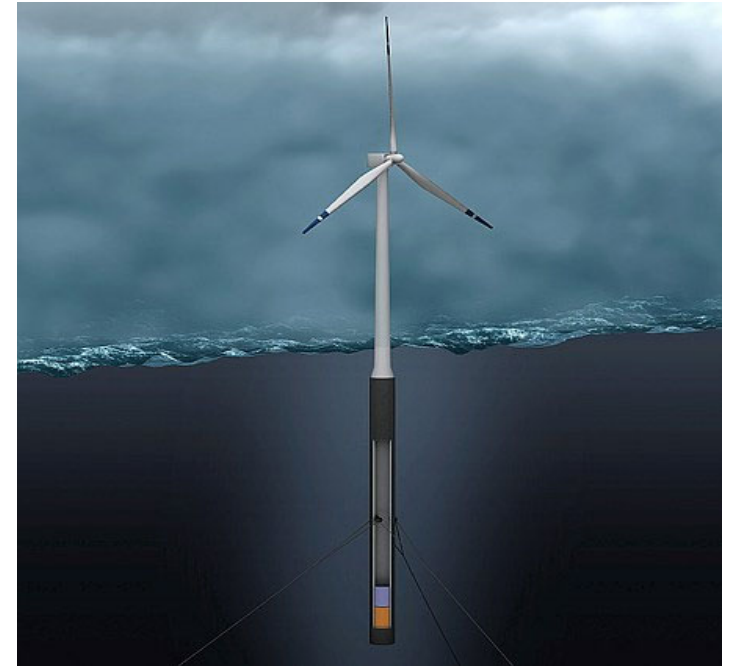


A simplified analysis of a semi-submersible, cont'd

- Surface piercing members / columns
- Vertical force on bottom is generally computed from Froude-Krylov force and added mass force (neglect diffraction)

$$F_3^{Froude-Krylov} = p_d A_S$$

- However, added mass force is only included if the flow (physically) surrounds the column end



The end

- Work on Assignment 3

Appendix related to Assignment 3

- Neglecting damping, the equation for vertical motion of the semi-sub reads, see **slide 12**:

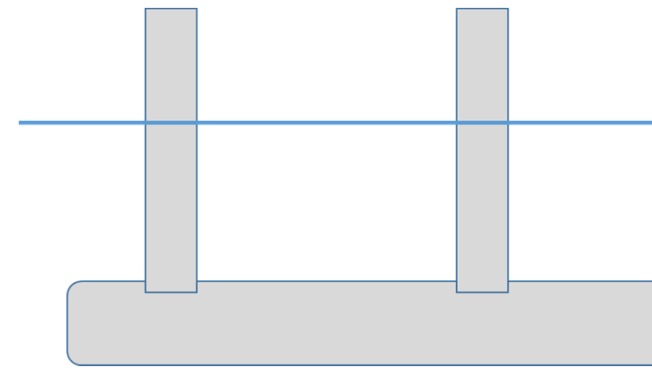
$$(\rho \nabla + a)\ddot{z} + cz = F_3(t)$$

which you must solve.

Suggested “recipe” and hints:

(1) If you have not done already, start by reading Chapter 15 in the lecture notes (Nielsen, 2024)

(2) Hints...



Appendix related to Assignment 3, cont'd

HINTS:

- Recall that the excitation force is computed for the single members of the semi-sub
- pontoons are fully submerged and we make use of the Morison formula to compute the “in-line (vertical) force” associated with vertical water particle motion resulting from regular waves. NB: You must consider the horizontal distance between the pontoons.
- For the legs (columns) there is only the contribution from the Froude-Krylov force, but you must again consider the spacing between the sets

